

SMART INDIA HACKATHON 2026

CampusMate AI

Smart Campus Navigation & Student Assistant

One intelligent platform for campus navigation, student assistance, and campus transport.

PROBLEM STATEMENT ID

SIH-2026-CAMPUSMATE

PROBLEM STATEMENT TITLE

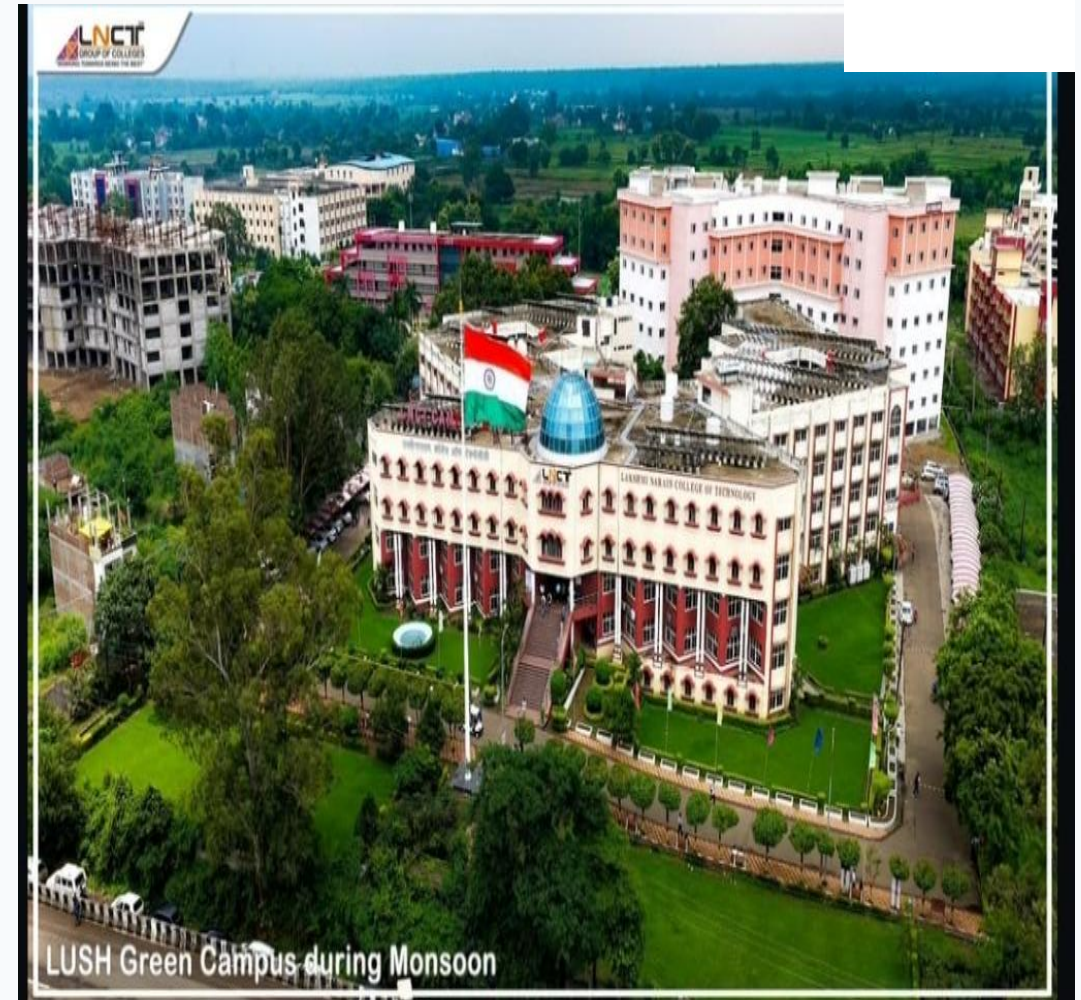
AI-Powered Smart Campus Navigation & Transport Companion

THEME & CATEGORY

Smart Education / Institutional Systems | Software

TEAM ID & TEAM NAME

9826585201 | Barbie Squad



Proposed Solution

Traditional Campus Challenges

- Information Fragmentation: Bus schedules, directories, and notices are scattered across static PDFs and bulletin boards.
- Navigation Hurdles: Visitors and new students struggle to find lecture halls, admin blocks, and facilities inside massive campuses.
- Transit Uncertainty: Lack of active timetable tracking or simulated progress leads to bus stop waiting anxiety.
- Support Delays: Simple academic queries require physical registration desk visits, causing time inefficiencies.

Proposed Unified Platform Solution

- Smart Navigation: 3D vector map with route guidance to major buildings.
- Simulated Bus Tracking: Stop timelines, shift filters, and live ETAs.
- Contextual AI Assistant: Multi-branch query responses backed by data.
- Explore Gallery: Integrated photo catalog mapping assets to blocks.



Technical Approach

User Interface Layer (Mobile-First Viewports)

HTML5 | Vanilla CSS | HSL Theme Colors | Responsive Breakpoints (320px-430px)

Frontend Interface & Module Controllers

- MapPage: Vector layouts, path highlighting, and photo overlay triggers
- BusTracking: Decoupled simulation clock, progress calculations, and ETAs
- Departments: Tabbed academic directories with search indices

Mathematical & Local Logic Services

- Haversine Distance Formula: Dynamic telemetry distance in kilometers
- Simulation Clock Engine: Progression intervals for mockup GPS coordinates

Factual Structured Database Layer (Local JS Modules)

- campusData.js: Location aliases, verified stop grids, schedules
- academicData.js: 17 schools and their respective course lists
- **Future Live GPS: Hardware GPS -> REST APIs -> Client WebSockets**

Verified Technology Stack

Languages

ES6+ JavaScript, HTML5, CSS3

Frontend Library & Build Tools

React 19 (Hooks, Context), Vite 8 (Hot Module Replacement)

Map Library & Tiles

Leaflet JS 1.9.4 | OpenStreetMap Tile Servers

Linting & Code Quality

Oxlint Linter (Zero warnings, zero errors compile)

Data Architecture

Local client JS data modules for instant, offline-ready queries

Feasibility & Viability

Feasibility Assessment

Technical Feasibility

- Lightweight client-side build requires no heavy database servers.
- Leaflet maps run smoothly on budget mobile processors.

Operational Feasibility

- Transport coordinators can update bus schedules easily via campusData.js.
- Standardised UI is user-friendly for all university stakeholders.



Risk & Mitigation Matrix

1. Inaccurate timing ETAs due to traffic

- Mitigation: Haversine distance telemetry shows actual physical distances.

2. GPS loss inside structures/basements

- Mitigation: Fallback stop-to-stop simulation clock interpolation.

3. Offline access constraints

- Mitigation: Pre-bundled local JS data runs without network connection.

Platform Limitations

- OpenStreetMap tiles require active internet connection to load.
- Real-time GPS tracking is simulated; hardware device installs on buses are required for live telemetry deployment.

Impact & Benefits

Ecosystem Benefits

Students

- Zero transit anxiety, immediate room directions, fast academic search.

Administration

- Modernised digital catalog, simplified student onboarding.

Transport Team

- Operational timing checks and clear stop-level visualizations.

Unique Value Proposition

- Replacing scattered PDF timetables, static maps, and directories with a single, mobile-responsive campus companion.



Credible Feature Comparison Matrix

Feature Capability	Generic Maps	Web Portals	CampusMate AI
Building Search	limited	no	Yes
Bus Tracking	no	static list	Yes
Timetable ETAs	no	static PDF	Yes
Contextual AI	no	no	Yes
Explore Gallery	no	limited	Yes
Mobile Layout	Yes	limited	Yes

Research & References

Bibliographical References

1. Institutional Information

- LNCT University Transport Department stop timing datasets.
- LNCT 360 Degree Virtual Tour directories and maps.

2. Technical Specifications & APIs

- Leaflet API documentation for markers, routing, and layer controls.
- OpenStreetMap Foundation map tile rendering server guidelines.
- React 19 component lifecycle rendering paradigms and Vite documentation.

3. Mathematical Standards

- Haversine formula specifications for geographic telemetry over spheres.

Project Verification Links

GitHub Remote Repository

- https://github.com/NidhiRajput06/Campusmate_ai

Active Local Wi-Fi Server Link

- <http://192.168.1.9:5173/> (Exposed on Wi-Fi for mobile)

